Elite Family Networks and the Reproduction of Political Power in Latin America

Evidence from Wikipedia Genealogical Data,
Nineteenth to Twenty-First Centuries

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Abstract.

How do elite family networks structure the reproduction of political power in Latin America from the nineteenth to the twenty-first century? We approach this as a problem in computational text analysis: Spanish-language Wikipedia is treated as a collectively authored biographical archive, from which we recover a multi-country kinship network through infobox extraction and hyperlink-based entity resolution (8,201 matched edges across ten republics), augmented by institutional layers for party affiliation, educational trajectories, and political succession.Across complementary analyses — stochastic block modeling, exponential random graph models, homophily decomposition, temporal cohort analysis, Gould–Fernandez brokerage decomposition, and targeted-attack simulations — we find that surface heterogeneity across republics coexists with a common generative architecture: an asymmetric core–periphery structure that is historically consolidated, geographically concentrated, and dependent on a small set of robust brokers to remain connected.We interpret the result as a structured description of a "documentary elite" encoded in Wikipedia — not an exhaustive map of Latin American elites, but a large-scale archive whose internal structure is sociologically informative in its own right. The paper is a first-pass demonstration of what coordinated corpus construction and common network metrics can reveal about elite reproduction when applied at scale across ten republics.

All project materials are publicly available at https://github.com/matdknu/familiaR-wiki.

Keywords: elite networks; Latin America; family; political reproduction; Wikipedia; social network analysis

1 Introduction

Latin American political history has long been interpreted through the lens of oligarchy, caudillismo, and the intergenerational transmission of status (Mahoney, 2010; Centeno, 2002). Classical elite theory—from Pareto and Mosca through mid-twentieth-century political sociology—stressed circulation versus closure: whether ruling strata renew themselves through co-optation and alliance or seal themselves behind kinship and class barriers. Empirical adjudication at scale, however, has remained difficult. Systematic, comparable evidence on *how* elite families connect across countries and centuries is scarce because prosopographical reconstruction is labor-intensive and rarely harmonized across national archives.

Historical sociology and political sociology nevertheless offer rich idiographic traditions. Padgett and Ansell's (1993) analysis of Florentine marriage and banking ties showed how multiplex network structure could sustain "robust action" under factional rivalry. Regional studies of Latin American *familias notables* documented marriage strategies linking land, office, and credit (Balmori, Voss, & Wortman, 1984). Scaling such insights to multiple republics typically required either intensive archival reconstruction or illustrative genealogies whose representativeness is uncertain.

The computational turn in historical network research has opened a second route. Over the past two decades, large-scale digital prosopographies have been assembled for Song-dynasty China (Bol, 2012), the Byzantine Empire (PBE project), medieval English elites, and twentieth-century European business networks (Bühlmann, David, & Mach, 2012), combining factoid-model databases with social network analysis to identify structural patterns that purely qualitative inspection cannot reach (Verboven, Carlier, & Dumolyn, 2007). Yet comparative elite research on Latin America has largely remained within the qualitative-archival tradition established by Balmori et al. (1984), Zeitlin (1984), and Stabili (1996): rich in ethnographic and documentary texture, but concentrated on single countries, short time windows, or small curated genealogies. To our knowledge, no prior study applies large-scale computational prosopography and social network analysis to Latin American elite kinship across multiple republics — neither using Wikipedia nor comparable digital sources. This paper sits in that gap. It is a first-pass computational-text contribution: the goal is not to replace qualitative historiography, but to show what a coordinated corpus and common network metrics can reveal across ten republics when analyzed at scale.

Digital traces offer a complementary path. Spanish-language Wikipedia aggregates biographical entries produced by dispersed editors; kinship statements—spouses, parents, children, partners—are often included in infoboxes or narrative leads. These data are neither representative of whole populations nor free of bias: they disproportionately document literate, urban, politically and culturally visible individuals. Yet precisely that bias may approximate what successive generations of national publics chose to record about "notable" families—a *documentary elite* whose network properties are sociologically informative in their own right (Lazer et al., 2014). The task is therefore not to treat Wikipedia as a mirror of society, but as a large, public, and evolving archive whose internal structure can be analyzed.

This paper asks: **How have elite family networks structured the reproduction of political power in Latin America from roughly the nineteenth through the twenty-first centuries?** We articulate five hypotheses. H1–H4 examine observable network features: (i) the association between marriage closure and political concentration, (ii) the alignment between macro-scale graph topology and stylized regime types, (iii) the geography of transnational kinship ties, and (iv) the identification of brokerage positions via betweenness centrality. H5 introduces an integrative perspective: it evaluates whether these heterogeneous surface patterns are consistent with an underlying structural regularity—specifically, a core–periphery architecture that is historically consolidated (not atemporal) and geographically asymmetric (not pan-Latin-American), and that jointly produces both rigidity and fragility in the network.

Our contribution is threefold, but rests on a single organizing claim: **family network structure is not merely a byproduct of political order in Latin America—it is one of the infrastructures through which that order is reproduced.** Where elites built durable states, they did so by layering kinship closure onto institutional access; where states remained fragile or personalist, the configuration of kinship ties reflects—and potentially constrains—those outcomes. We do not claim causal identification. Rather, we show that network topology *covaries systematically* with regime characteristics across ten republics using a common corpus and comparable metrics—an empirical contribution that large-scale comparative prosopography has rarely achieved.

We organize the analysis around three conceptual distinctions. First, we distinguish **closure** from **reach** as alternative elite network strategies: families that maximize endogamy (closure) tend to sacrifice bridging ties, while those that expand across borders typically exhibit lower endogamy rates. This **closure–reach trade-off** is observable in degree–endogamy correlations and is theoretically grounded in the tension between Bourdieu's (1996) reproductive closure and Granovetter's (1973) structural holes.

A corollary of this trade-off is that no single family should simultaneously maximize both dimensions. However, under a core–periphery configuration, a small subset of families may approximate this dual position: they are embedded in a dense core (high closure at the block level) while also maintaining bridging roles across blocks (high reach at the graph level). We refer to this configuration as **robust brokerage**, following Padgett and Ansell (1993). H5 evaluates whether such families are empirically detectable, how frequent they are, and what structural role they occupy.

Second, we distinguish the **documentary elite field**—the subset of families that are consistently recorded and interlinked in public digital archives—from the broader social elite. This distinction is central: the documentary elite field is not a proxy for the entire elite, but a specific object of analysis whose boundaries and biases are themselves sociologically meaningful.

Third, we distinguish between **surface heterogeneity** and **underlying structural regularity**. While H1–H4 document substantial variation across countries in observable network features, H5 assesses whether these differences coexist with a common structural backbone. The aim is not to collapse heterogeneity into a single model, but to evaluate whether diverse configurations are compatible with a shared underlying architecture.

The remainder proceeds as follows. Section 2 reviews relevant literature, including core–periphery and robust-brokerage frameworks that motivate H5. Section 3 describes data construction, variables, and limitations—with summary statistics reported in the descriptive tables. Section 4 presents H1–H4 results. Section 4.5 presents the integrative H5 analysis, combining ERGM, homophily decomposition, stochastic block models, temporal cohort analysis, and targeted-attack simulations as complementary (though not independent) approaches to characterizing network structure. Section 5 discusses interpretation, scope conditions, and limitations, and concludes.

2 Literature and theoretical background

2.1 Notable families, oligarchy, and the state

Classic scholarship on Latin America emphasized the *familia notable* as a unit linking land, urban property, marriage, and access to state office (Balmori et al., 1984). Centeno (2002) connected elite networks to war-making and the fiscal–military organization of republics, while Mahoney (2010) emphasized path dependence in colonial and postcolonial institutions. Together, these traditions imply testable expectations at the network level: **family strategies**—endogamy, strategic exogamy, and placement of kin in parties and ministries—should covary with the concentration of political access and the durability of elite coalitions over time.

Bourdieu's (1996) analysis of the French state nobility provides a vocabulary for **social reproduction**: elites convert economic resources into social ties and cultural credentials, then reproduce privileged access to institutional positions across generations. Although Latin American contexts differ—through the roles of the Catholic Church, the military, export agriculture, and mining enclaves—the underlying problem of stabilizing advantage through kinship remains analogous. Where elites successfully restrict marriage markets, one expects measurable **closure** at the level of surname clusters (H1). Where political competition fragments access, closure may coexist with relatively dense but outward-oriented networks.

2.2 Network topology and political orders

Historical network analysis demonstrates that **macro-structure**—chains, stars, dense cores, or disconnected fragments—can correspond to distinct modes of coordination and conflict (Padgett & Ansell, 1993). Granovetter (1973) motivated attention to weak ties and brokerage across clusters. Translating these ideas to Latin America requires caution: one may *loosely* associate linear dynastic chains with personalist or patrimonial consolidation (in Weberian terms), star-like graphs with legitimating "founding" lineages, dense webs with competitive oligarchies and multiple channels of access, and fragmented graphs with regionalized or weakly integrated elites. These mappings function as **interpretive ideal-types**, not as deterministic classifications inferred mechanically from graph metrics (H2).

2.3 Empire, corridor, and transnational elites

Colonial geography—mining belts, highland–coastal corridors, and viceregal administrative centers—may channel later elite mobility and marriage patterns. Pacific Andean spaces, along with Chilean–Peruvian histories of war and diplomacy, plausibly leave **residual structure** in cross-border elite ties long after independence. If transnational edges in a Wikipedia-derived kinship graph concentrate along a Pacific–Andean corridor rather than dispersing uniformly, that pattern provides descriptive support for **path-dependent geography** (H3). Such evidence should be interpreted as consistent with historically informed priors, rather than as direct identification of colonial causal mechanisms.

2.4 Gender, marriage, and visibility

Colonial and republican marriage regimes structured women's roles as alliance partners and transmitters of property and status (Lavrin, 1990). If encyclopedic records disproportionately document women through ties to notable husbands and sons, **degree** in kinship-based graphs may overstate public or political centrality, while **betweenness** in multiplex projections remains comparatively lower. We treat such gendered divergences as **hypothesis-generating patterns** about brokerage and exclusion, rather than as definitive evidence of underlying mechanisms, given the limitations of name-based or archival proxies.

2.5 Core–periphery architectures and robust brokerage

Beyond topological ideal-types, network scholarship has formalized structural configurations that are directly relevant to elite coordination. *Core–periphery* models (Borgatti & Everett, 2000) identify graphs in which a small, densely interconnected group is embedded within a larger, sparsely connected periphery, with intermediate tie density between them. Such architectures are consistent with elite systems in which a coordinating stratum maintains internal cohesion while sustaining selective outward connections. Detecting these patterns typically relies on unsupervised approaches such as stochastic block models (SBM), which infer block structure from observed tie distributions.

Robust brokerage, as developed by Padgett and Ansell (1993), refers to actors that are simultaneously embedded within a dense core and positioned to bridge across its boundaries. This dual position enables what they term *robust action*: the capacity to maintain flexibility across domains while preserving stability within them. Operationally, this configuration can be evaluated using the Gould and Fernandez (1989) typology of brokerage roles—coordinator, consultant, gatekeeper, representative, and liaison. Actors occupying robust brokerage positions are expected to exhibit relatively high shares of cross-block roles (e.g., gatekeeper, representative) alongside strong within-core embedding.

H5 evaluates whether the empirical patterns described in H1–H4 are **consistent with** such a structural configuration—namely, an asymmetric core–periphery architecture in which a limited number of families approximate this dual role. Rather than positing a single generative model, the aim is to assess whether multiple descriptive indicators converge on a common structural interpretation within the documentary elite field.

3 Data and methods

3.1 Source, pipeline, and corpus construction

3.1.1 Methodological framing

We treat Spanish-language Wikipedia as a **collectively authored biographical archive** and approach the construction of our corpus as a problem in *computational text analysis* (Grimmer, Roberts, & Stewart, 2022; Edelmann et al., 2020). The analytical object is not Wikipedia itself but the network of elite kinship that can be *recovered* from it by systematic processing. Recovery requires a pipeline that addresses, in sequence, three problems classical to computational social science when working with large text archives: *entity enumeration* (which biographical articles belong to the corpus?), *entity extraction* (what attributes and relations does each article ascribe to its subject?), and *entity resolution* (when does a name mentioned in one article refer to the subject of another article in the same corpus?).

Wikipedia's internal structure — the category system and the infobox convention — substantially pre-solves the first two problems. Editors have, over two decades, partitioned biographies of notable Latin Americans into `Categoría:Familias_de_<País>` taxonomies, and have standardized biographical summaries into infobox panels with consistent field vocabularies. A computational pipeline built on these structural regularities can extract a reliable, if bounded, documentary record. The third problem — resolving mentioned relatives to scraped biographies — is the core *non-trivial* step and is the one that determines the validity of the resulting graph.

The pipeline proceeds in four stages that convert Wikipedia's text-plus-structure archive into the multiplex network object analyzed in Section 4.

3.1.2 Stage 1 — Corpus boundary via the family-category taxonomy

The first problem is to define which biographies belong to the corpus. We delegate this decision to Wikipedia's editorial community: the corpus for each country consists of all biographies reachable by recursive descent through the subcategory tree rooted at `Categoría:Familias_de_<País>`. This choice substitutes **editorial sociology** for researcher-curated family lists. Its advantages are scalability, cross-country comparability under a uniform rule, and resistance to selection biases arising from the researcher's prior knowledge of which families "matter"; its principal cost is that Wikipedia's taxonomy reflects the documentary attention of a specific editorial community and will systematically under-represent elites whose public visibility is lower in Spanish-language digital discourse (Section 3.5). A family, in this paper, is therefore a Wikipedia category — a sociologically constructed grouping of biographies that some community of editors judged cohesive enough to merit a shared categorical label.

3.1.3 Stage 2 — Extracting entities from the biographical infobox

Wikipedia biographies contain both unstructured prose and a structured **infobox** panel summarizing the subject's vital statistics, career, and relational attributes. The infobox is a collectively enforced template: Spanish-language biographies of Latin American political figures share a near-standardized field vocabulary that identifies, for each subject, a surname (`familia`), parents, spouse and partner, siblings and children, political offices held, predecessor and successor in each office, and party and educational affiliations. These fields function as pre-annotated named entities: the editorial community has already performed, through template compliance, the entity-recognition task that would otherwise require supervised NLP.

Our extraction targets this structured layer only. This is a deliberate methodological choice. Free-text relation extraction from biographical prose is feasible but introduces substantial recall/precision tradeoffs and language-specific error modes that would complicate cross-country comparison. Restricting to the infobox yields a smaller but cleaner record: an individual enters the corpus with a defined surname, zero or more kin relations, and — when present — political office information including succession chains that will become the institutional layer of the multiplex graph (Section 3.3).

3.1.4 Stage 3 — Resolving names to biographies through the hyperlink graph

The step that converts textual kinship mentions into graph edges is entity resolution: when a biography of individual A names "Pedro Aguirre Cerda" as a parent, does that string refer to the subject of another biography in our corpus? The standard computational approach to this problem combines string similarity with disambiguation heuristics (Grimmer et al., 2022, ch. 12); we exploit a simpler and more precise mechanism available in Wikipedia specifically.

Editors encode disambiguation directly in the markup. When Wikipedia biography A names B as a relative and B has a dedicated biography, editors routinely render the name as an internal hyperlink to B's URL. This hyperlink — not the string name itself — is what our pipeline uses to resolve relations. An edge `A - B` enters the kinship graph if and only if both A and B are scraped biographies in our corpus and at least one names the other through such a link. String-level mentions without a resolving hyperlink are retained as attributes of the source individual but do not generate edges. This rule trades kinship recall for edge precision: **57.8% (8,201 of 14,190)** of infobox-declared kinship statements resolve to a second scraped biography and become graph edges. The remaining 42.2% name a relative who either has no Wikipedia biography, has one that lies outside the family-category tree for that country, or is named without a hyperlink in the source infobox.

The analogous resolution procedure applies to the institutional layers. Political-succession edges are drawn directly from the `predecesor` and `sucesor` infobox fields, both of which are typically rendered as internal links. Party and educational co-affiliation edges require an additional normalization step — two biographies listing "Partido Liberal" need to be matched as affiliated with the same entity — but the basic logic is unchanged.

3.1.5 Stage 4 — Aggregation and the working corpus

Per-individual records are consolidated across all families and countries into a normalized relational schema. Deduplication on canonical Wikipedia URL removes individuals who appear in multiple family categories (e.g., a woman listed under both her birth family and her husband's family), retaining the union of their attributes. The resulting working corpus contains **6,711 individuals** across ten countries, linked by **8,201 matched kinship edges** and by the additional multiplex layers described above. Of these individuals, **92.9%** carry a parsed birth year, which anchors the cohort analysis in Section 4.5.

3.2 Family, country, and measurement definitions

Family is operationalized via *familia_norm*, a normalized surname cluster constructed through orthographic harmonization. This construct should be interpreted as a **documentary grouping** derived from Wikipedia conventions rather than as a validated genealogical lineage.

Country is assigned using nationality fields, infobox cues, and text-based inference, complemented by manual verification where ambiguity arises. A kinship edge is classified as **transnational** when the assigned countries of its endpoints differ. Both family and country variables are therefore subject to classification error inherent to the source.

3.3 Network construction and metrics

We construct an undirected graph based on matched kinship relations and augment it with additional relational layers, including within-family ties for small lineages, predecessor–successor links, and affiliations derived from shared party membership and educational institutions. Together, these layers define a **multiplex network** used for centrality estimation and community detection.

Accordingly, degree and betweenness centrality are computed on the multiplex graph, whereas marriage-based endogamy measures rely exclusively on kinship edges. This distinction separates **institutional connectivity** from strictly **genealogical closure**.

3.3.1 Why betweenness is computed on the multiplex graph

Betweenness centrality in a kinship-only graph captures mediation along genealogical chains, an analytically narrow concept tied to parent–child and spousal links. The multiplex specification allows shortest paths to traverse institutional as well as familial relations—such as shared party affiliation, educational trajectories, and succession ties—thus approximating a broader **infrastructure of elite coordination**.

This specification does not imply that multiplex betweenness directly measures political influence; rather, it captures **structural positioning** within the observed relational system. By contrast, endogamy rates are computed exclusively on spouse and partner ties, preserving a strictly **matrimonial definition** of closure.

Network measures are computed with `igraph`, including degree, betweenness centrality, PageRank, local clustering coefficient, and community structure via modularity optimization. Additional aggregate indicators are constructed as follows:

- **Endogamy:** On spouse/partner edges with matched partners, a tie is classified as endogamous if both endpoints share `familia_norm`.
- **Transnational structure:** Country-pair aggregates combine kinship ties with biographical co-mention signals, which should be interpreted as complementary indicators of cross-border linkage.
- **Dynasty persistence:** Summarized through longitudinal patterns of surname continuity and connectivity.
- **Country-level summaries:** Aggregate indicators are reported to support hypothesis-oriented comparisons.

3.4 Analytical stance and temporal scope

All analyses are **observational**. Network measures describe co-occurrence patterns within a digital corpus shaped by notability and editorial practices. Throughout Section 4, we adopt a **descriptively cautious language** (e.g., “associated with,” “consistent with,” “suggestive”) to avoid causal over-interpretation.

Biographical dates span approximately **1800–2000**, with some colonial antecedents. Unless explicitly disaggregated, network representations **pool multiple historical periods**, implying that observed structures combine relationships formed under different institutional contexts.

3.5 Limitations

1. **Selection bias:** Wikipedia disproportionately documents literate, urban, and politically visible individuals; indigenous, Afro-descendant, and subnational elites are underrepresented.
2. **Kinship incompleteness:** Many familial ties are unrecorded; women may appear primarily as relational nodes (e.g., spouses or mothers).
3. **Label uncertainty:** *familia_norm* may merge distinct lineages or split unified ones, particularly across national contexts.
4. **Temporal aggregation:** Networks combine ties across historical periods unless explicitly segmented.
5. **Edge incompleteness:** A substantial share of kinship relations lack a matched endpoint, limiting full network reconstruction.
6. **Country assignment constraints:** Multinational biographies are reduced to a single country label, partially mitigated by biographical cross-referencing.
7. **Measurement dependence:** Multiple network-level metrics (e.g., centrality, block structure, robustness) are derived from the same underlying graph. They should be interpreted as **complementary perspectives on a common object**, not as statistically independent tests. The epistemic value of their convergence (Section 4.5) lies in the diversity of inferential assumptions each brings to bear, not in sample independence.
8. **No causal identification:** Network position measures (e.g., betweenness, endogamy) cannot be interpreted as causal drivers without additional research designs.

These limitations position the analysis as a **structured mapping of a documentary elite field**, rather than a comprehensive representation of Latin American elites. They also motivate the interpretive discipline maintained in Section 4 and Section 5.

4 Results

We organize the results by hypothesis. Summary statistics for marriage closure and institutional concentration by country appear in Table 1 and Table 2. Cross-border structure is documented in Table 3 and Table 4. Figure 1 situates the temporal distribution of births used throughout. National kinship layouts appear in Figure 2; transnational summaries follow in Figure 3.

4.1 H1: Family closure and political concentration

We begin with a descriptive overview of the corpus. The working corpus comprises **6,711 individuals** across ten countries, each recorded with a set of core variables — family of affiliation, country of birth, country of death, kinship relations (siblings, spouses, descendants, ascendants), and associated dates. Figure 1 displays the distribution of birth years by country: documentation density peaks around the middle of the nineteenth and twentieth centuries, reflecting both the expansion of republican political life and the uneven biographical coverage of earlier cohorts.

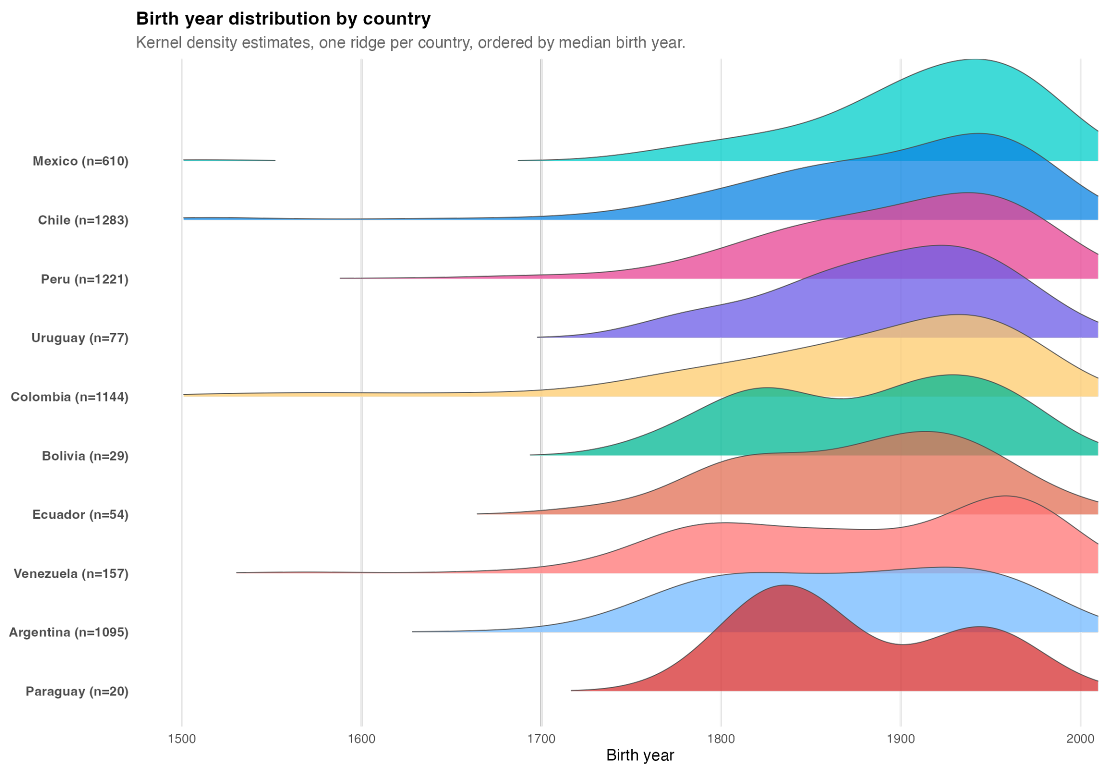

Figure 1. Birth year distribution by country (kernel density ridges; documented individuals with non-missing birth year).

Endogamy rates vary markedly across countries (Table 1). Among the larger republics, kinship-based endogamy rates cluster roughly between **0.77** and **0.87**, while **Paraguay** reaches **1.00** on a very small number of observed marriages (**4**). **Bolivia** shows the lowest endogamy rate in the table (**0.67**) alongside a high transnational marriage share (**0.67**), a pattern that likely reflects both small sample size and corpus boundary effects. Table 2 reports family counts by country (e.g., **Argentina 165**, **Peru 182**, **Mexico 111**, **Chile 98**) together with descriptive correlations between family-level closure and two network indicators: genealogical degree and multiplex betweenness. Cells marked "—" indicate estimates not reported because of insufficient numbers of families.

Table 1. Marriage records and endogamy rates by country (matched partners; kinship-based definitions).

Country	Marriages	Endogamous	Endogamy rate	Transnational marriages	Transnational rate
Paraguay	4	4	1.000	0	0.000
Ecuador	10	9	0.900	2	0.200
Mexico	108	94	0.870	15	0.139
Argentina	177	153	0.864	36	0.203
Venezuela	25	21	0.840	5	0.200
Chile	251	210	0.837	23	0.092
Colombia	151	120	0.795	13	0.086
Uruguay	18	14	0.778	7	0.389
Peru	104	80	0.769	12	0.115
Bolivia	3	2	0.667	2	0.667

At the country level, **Corr(degree, closure)** is negative wherever it is defined in Table 2, and substantively large in several major republics: **Mexico −0.797**, **Venezuela −0.764**, **Argentina −0.738**, **Peru −0.720**, **Chile −0.536**, and **Colombia −0.478**. One observational reading—consistent with competitive-oligarchy narratives and with Padgett and Ansell's (1993) account of robust action—is that families maintaining broad network reach may do so partly by forgoing high levels of matrimonial closure. Larger and more connected lineages may therefore face a trade-off between preserving closure within the surname cluster and sustaining broader alliance networks. Smaller or less central families, by contrast, may maintain higher closure without sacrificing as much documented connectivity.

Table 2. Country-level family counts and descriptive correlations between genealogical degree and family closure (paper summary table).

Country	Families (n)	Median party conc.	Median office conc.	Corr(degree, closure)	Corr(betweenness, closure)
Argentina	165	0.500	0.500	-0.738	-0.236
Bolivia	4	0.417	0.550	—	—
Chile	98	0.375	0.529	-0.536	-0.330
Colombia	147	0.667	0.556	-0.478	-0.073
Ecuador	6	0.700	0.472	—	—
Mexico	111	0.667	0.667	-0.797	-0.440
Paraguay	1	—	1.000	—	—
Peru	182	0.500	0.571	-0.720	-0.122
Uruguay	9	0.833	0.615	—	—
Venezuela	20	0.500	0.625	-0.764	-0.344

Note. Correlations are not reported for countries with fewer than 10 families

Read through Bourdieu's vocabulary of capital conversion, brokerage across families and institutions and closure within the surname cluster appear here not as mechanically complementary dimensions, but as partly competing strategies of elite reproduction. The pattern remains descriptive: it identifies a regularity in the documentary corpus rather than a causal mechanism.

Corr(betweenness, closure) is more modest in magnitude than **Corr(degree, closure)** in every country where both are defined—for example **Argentina −0.236**, **Mexico −0.440**, **Chile −0.330**, **Venezuela −0.344**, **Peru −0.122**, and **Colombia −0.073** (Table 2). This suggests that structural brokerage in the multiplex graph is less tightly organized around matrimonial closure than raw genealogical degree is. The pattern is consistent with shortest paths that traverse institutional layers—party affiliation, educational trajectories, and succession ties—that need not align with endogamy rates computed on spouse and partner edges alone.

This distinction matters substantively. It suggests that two classic mechanisms of elite reproduction—marriage strategy and institutional placement—may operate with partial independence. Families can remain relatively closed in matrimonial terms while some of their members still occupy brokerage positions in institutional space, and vice versa. In that sense, **H1 receives partial descriptive support**: closure and institutional concentration co-vary in informative ways, but the association is neither uniform across countries nor sufficient on its own to imply a single model of elite reproduction.

4.2 H2: Topology and stylized regime types

The country-level layouts in Figure 2 suggest recurrent but heterogeneous topological patterns. **Paraguay** appears close to a linear chain centered on the López dynasty, evocative of patrimonial consolidation. **Chile** and **Peru** display more hub-centered configurations, with prominent nodes connecting multiple branches. **Argentina**, **Colombia**, and **Mexico** show denser webs with several high-degree families, while **Bolivia**, **Venezuela**, and **Ecuador** appear more fragmented into smaller cohesive clusters.

These associations are interpretive rather than algorithmic. The argument is not that graph shape mechanically encodes regime type, but that certain recurrent forms are broadly consistent with stylized historical narratives of personalist consolidation, oligarchic competition, or weakly integrated elite fields. Figure 2 therefore supports **H2 at the level of ideal-typical analogy**, not predictive classification.

4.3 H3: Transnational corridors

Figure 3 maps the aggregate geography of cross-border elite linkages: edge widths encode total tie counts between country pairs, and node sizes reflect each country's overall transnational flow strength. The resulting pattern is strongly structured rather than uniform — dense along a Pacific–Andean corridor and across the River Plate, markedly thinner elsewhere.

Figure 2. Family kinship networks by country (exploratory faceted layouts; node size encodes degree in each subgraph).

Table 3. Strongest country–country ties (kinship + biographical signals), top 15 pairs by total weight.

From → To	Kinship	Biographical	Total
chile → peru	36	81	117
argentina → uruguay	46	59	105
argentina → chile	56	38	94
argentina → peru	31	59	90
argentina → colombia	30	27	57
chile → colombia	43	14	57
colombia → peru	23	28	51
colombia → venezuela	37	11	48
chile → mexico	17	27	44
argentina → mexico	19	7	26
argentina → bolivia	12	13	25
colombia → ecuador	13	9	22
colombia → mexico	12	7	19
mexico → peru	6	9	15
argentina → paraguay	7	5	12

At the family level, Table 4 highlights lineages such as **Borrero** (**39** members across **four** countries: Argentina, Chile, Colombia, Ecuador), **Campero** (**24** members; Argentina, Bolivia, Chile, Peru), and **Sucre** (**11** members; Argentina, Colombia, Ecuador, Venezuela).

Their geographic footprints are consistent with post-independence elite continuity across Bolivarian and Gran-Colombian spaces, though again this should be read as interpretive alignment rather than causal proof.

Table 4. Multinational families ranked by number of countries represented (top 12 by breadth × size).

Family	Countries	Members	Country list
Borrero	4	39	argentina, chile, colombia, ecuador
Campero	4	24	argentina, bolivia, chile, peru
Ponce De León	4	12	argentina, mexico, peru, uruguay
García De Zúñiga	4	11	argentina, chile, colombia, uruguay
Sucre	4	11	argentina, colombia, ecuador, venezuela
Pinto Chile	3	50	argentina, chile, uruguay
Caicedo	3	48	chile, colombia, peru
Velasco Chile	3	44	argentina, chile, venezuela
Diez Canseco	3	33	chile, ecuador, peru
Pizarro	3	33	chile, colombia, peru
Miró Quesada	3	22	mexico, peru, uruguay
O'donnell	3	22	argentina, colombia, mexico

Aggregated weights rank **Chile–Peru** first (**117** total counted events), followed by **Argentina–Uruguay** (**105**), **Argentina–Chile** (**94**), and **Argentina–Peru** (**90**) (Table 3; Figure 3). Atlantic-oriented pairs such as **Argentina–Brazil** are absent from the top of this list in the current aggregation.

The decomposition into kinship versus biographical counts matters theoretically. **Chile–Peru** combines **36** kinship and **81** biographical events—roughly **31%** kinship and **69%** biographical—so the strongest dyad in the table is driven primarily by biographical co-mention rather than by cross-border genealogy alone. By contrast, **Argentina–Chile** shows the reverse balance: **56** kinship and **38** biographical events, making it the most genealogically entangled pair among the top ranks. **Argentina–Uruguay** is more mixed (**46** versus **59**), consistent with a dense Rioplatense elite field in which family and professional circulation are both heavily documented. These contrasts are descriptive: kinship-heavy pairs indicate actual cross-border family linkage in the observed graph, whereas biographical-heavy pairs capture transnational circulation that may or may not coincide with kinship.

Historical narratives are suggestive rather than identified here. The former **Viceroyalty of Peru** linked Chilean and Peruvian territories administratively, while the **War of the Pacific (1879–84)** plausibly intensified biographical cross-referencing between Chilean and Peruvian entries. **Argentina–Chile** is consistent with long-standing Andean–Southern Cone mobility, including nineteenth-century political exile, while **Argentina–Uruguay** fits a broader River Plate elite field. The absence of a strong **Argentina–Brazil** edge—despite the scale and proximity of both countries—is consistent with the idea that documentary elite ties are shaped more strongly by administrative corridors and historical circuits than by proximity alone.

The Pacific-corridor pattern is not self-explanatory. At least three mechanisms are compatible with it. At the *colonial* scale, viceregal administration linked legal institutions, ecclesiastical hierarchies, and commercial circuits across Pacific-Andean territories. At the *republican* scale, warfare, diplomacy, and exile may have intensified biographical entanglement independently of genealogy. At the *demographic* scale, highland urban centers such as Lima, Santiago, La Paz, and Bogotá may have constituted relatively integrated marriage markets among literate notables.

The continental view in Figure 3 makes this geographic structure legible: the Pacific–Andean spine (Chile–Peru–Colombia, with Argentina as a southern anchor) carries the heaviest edges, while Central America and the Atlantic coast appear comparatively thin. The figure should be read as a descriptive map of the documentary corpus, not as an exhaustive cartography of Latin American elite ties.

None of these mechanisms is identified here. Each remains a hypothesis for future cohort-level analysis, archival validation, or quasi-experimental designs. Even so, **H3 receives descriptive support** for a Pacific–Andean and Southern Cone emphasis in this documentary corpus, with the crucial caveat that biographical layers inflate some dyads relative to kinship and that mechanisms remain open.

4.4 H4: Betweenness and brokerage

High betweenness is concentrated among a relatively small set of individuals (Table 5), including **Luis Felipe Villarán** and **Walter Humala** in **Peru**, **José Evaristo Uriburu** in **Argentina**, and **Cuauhtémoc Cárdenas** in **Mexico**. More substantively, the specific figures that occupy these positions align with well-documented historical roles: Villarán and Cisneros anchor late-nineteenth and twentieth-century Peruvian institutional networks; Uriburu, Obligado, and Sáenz Peña are canonical members of the Argentine oligarchic "Generation of 1880"; and the continued presence of Di Tella and Cárdenas indicates brokerage persistence into twentieth-century party politics. In that limited sense, **H4 is descriptively supported**: brokerage positions are not only identifiable but also map onto historically plausible coordinators. We do not equate multiplex betweenness with real-world political influence outside the documented network.

Table 5. Top 15 individuals by betweenness centrality (multiplex graph).

Name	Family	Country	Degree	Betweenness
Luis Felipe Villarán	villarán	Peru	410	517,714
Walter Humala	humala	Peru	557	473,134
Luis Jaime Cisneros	cisneros	Peru	364	391,641
José Evaristo Uriburu	álvarez de arenales	Argentina	398	328,970
Pastor Obligado	obligado	Argentina	324	267,974

Guido Di Tella	di tella	Argentina	276	240,510
Patrocinio González Garrido	ortiz mena	Mexico	194	212,889
Francisco Uriburu (hijo)	uriburu	Argentina	313	208,067
Cuauhtémoc Cárdenas	cárdenas	Mexico	266	198,005
Demetrio Sodi de la Tijera	sodi	Mexico	169	188,638
Roque Sáenz Peña	sáenz peña	Argentina	263	168,155
Guillermo Fernández de Soto	camacho	Colombia	219	163,071
Jaime Arrubla Paucar	arrubla	Colombia	172	146,224
Rafael Roncagliolo	orbegoso	Peru	235	144,958
Tomás Lander	lander	Venezuela	177	144,923

The concentration of top betweenness scores among Peruvian and Argentine figures invites a structural reading. Peru's prominence in the ranking is consistent with Lima's historical role as a hub linking Andean, coastal, and institutional circuits. Likewise, Argentina's high-betweenness cluster—Uriburu, Obligado, Di Tella, Sáenz Peña—is consistent with the dense layering of kinship, party, and educational ties often associated with the oligarchic "Generation of 1880."

These interpretations should remain cautious. High betweenness in the multiplex graph reflects structural location within the observed relational system, not intrinsic influence or historically verified brokerage in every case. Still, the distribution is consistent with the claim that some individuals occupy positions that bridge otherwise separated institutional and genealogical domains.

4.5 H5: Core–periphery architecture, historical consolidation, and systemic fragility

H1–H4 document observable properties of the elite network—closure, topology, transnational corridors, and brokerage. H5 evaluates whether these patterns are consistent with a deeper structural organization. Specifically, we assess whether the network exhibits a core–periphery architecture that is historically consolidated and structurally fragile. The analysis integrates multiple complementary approaches that converge on a common interpretation.

4.5.1 Tie formation

The network is extremely sparse at baseline, indicating that kinship ties are rare relative to the number of possible family pairs. At the same time, tie formation is strongly structured. Connections are substantially more likely within countries than across them, indicating robust national homophily operating alongside the transnational corridors documented in H3.

Family size also contributes to tie formation: larger lineages tend to accumulate more connections, suggesting a mild preferential attachment process. At the same time, ties are less likely between families with very different levels of connectivity, indicating stratification

Figure 3. Continental view of cross-border elite linkages (aggregated transnational flow map, pooled across all birth cohorts).

by network position. Taken together, these patterns describe a system that is both sparse and strongly sorted (Table 6).

Table 6. Exponential random graph model: tie-formation parameters on the family-level kinship graph.

Term	Estimate	Std. error	Odds ratio	p-value
edges	-8.776	0.160	0.000	$< 1e\text{-}300$
nodecov.n_members	0.056	0.002	1.057	2.5e-115
nodematch.pais	1.781	0.134	5.937	2.4e-40
absdiff.mean_degree	-0.352	0.117	0.703	0.0025

4.5.2 Homophily

Sorting operates differently across attributes. Structural position shows near-perfect homophily: families tend to connect to others with similar levels of connectivity. Country exhibits substantial but incomplete homophily, while temporal proximity plays a weaker role.

Table 7. Attribute-level homophily indices, with within-share and permutation p-values.

Attribute	H (observed)	H (expected)	Within-share	p-value
Country	0.479	0.189	0.578	<1e-300
PoliticalOffice	-0.855	0.879	0.776	<1e-300
NetworkTier	1.000	0.545	1.000	<1e-300
Era	0.234	0.505	0.621	<1e-300

By contrast, political roles display strong heterophily. Families do not primarily connect to others occupying the same institutional position; instead, ties are more common across roles. This pattern is consistent with functional complementarity: elite families specialize in different institutional domains while remaining embedded within the same structural and national fields.

4.5.3 Latent structure

The SBM operates on the family-level aggregation of the kinship graph (n = 743 families across the ten countries), not on the individual-level graph used elsewhere.

The SBM, applied without supervision, selects **K = 2** blocks (ICL-minimum). The two blocks are radically asymmetric: Block 1 contains **722** families with an internal tie probability of **0.00066**, while Block 2 contains **21** families with an internal tie probability of **0.138** — a ratio of roughly **208 to 1** (Table 8). Between-block ties occur at an intermediate rate ($\pi_{12} = 0.0020$) that is **three times** higher than within the periphery but **roughly seventy times** lower than within the core. Because the underlying graph is undirected, the block-pair probabilities are symmetric ($\pi_{12} = \pi_{21}$); both orderings are reported in Table 8 for readability. The model does not detect two rival coalitions; it detects a **core–periphery** structure in which a small, densely interconnected group sits above a much larger, sparsely connected field.

Table 8. Stochastic block model: inferred block sizes and within/between-block tie probabilities. The graph is undirected; cross-block rows are symmetric.

Block pair	Tie probability
B1 – B1	6.6e-04
B1 – B2	2.0e-03
B2 – B1	2.0e-03
B2 – B2	1.4e-01

The twenty-one families of the core are **geographically concentrated**: sixteen are Chilean, four are Colombian (Acevedo, Caicedo, Eder, Obregón), and one is Mexican (Pinal Hidalgo). No families from Argentina, Peru, Mexico (except Pinal Hidalgo), Venezuela, or the smaller republics enter the core. This is **striking** given that Peru contributes the largest national sub-corpus (**185** families) and Argentina the second-largest (**161**). **Sub-corpus size does not predict core membership**; structural position in the kinship graph does. We return to the interpretation of this asymmetry — and to its potential reading as a sampling artifact of Spanish-language editorial practices — in Section 5.

4.5.4 Historical trajectory

The core–periphery architecture is not atemporal. Splitting the graph by birth cohort (Table 9) reveals a monotonic decline in the share of ties that cross family boundaries: **17.7%** in the Colonial cohort (pre-1800) falls to **9.9%** at Independence (1800–1850), **6.7%** in the Oligarchic period (1850–1900), **5.5%** in the Republican era (1900–1950), and **5.3%** in the Modern cohort (1950–2000) — a **70% decline** across five successive cohorts. The decline is not driven by a collapse in edge count: edge volume *increases* from **1,658** (Colonial) to **2,366** (Republican) before falling in the Modern period. The reduction in the Modern cohort edge count (**868**) partly reflects right-censoring — the twentieth-century end of the corpus clips descendants — and should be read with that qualification.

Table 9. Network properties by birth cohort.

Cohort	Edges	Density	Cross-family share	Cross-country share	Gini (degree)
Colonial (pre-1800)	1658	2.4e-03	0.177	0.086	0.338
Independence (1800-1850)	1714	2.0e-03	0.099	0.042	0.306
Oligarchic (1850-1900)	1556	1.8e-03	0.067	0.030	0.269
Republican (1900-1950)	2366	1.3e-03	0.055	0.043	0.283
Modern (1950-2000)	868	3.0e-03	0.053	0.047	0.251

This conjunction — rising edge volume with falling cross-family share — is the signature of **oligarchic consolidation**. The network does not become smaller; it becomes *more self-referential*. Colonial elite networks spread ties across many lineages at low density; nineteenth- and early-twentieth-century networks concentrated a growing volume of ties within fewer lineages. The pattern is consistent with Centeno's (2002) thesis that the Latin American nineteenth century produced not strong states with autonomous bureaucracies but **oligarchic states** in which kinship intensification substituted for fiscal-military institution-building. The cross-country share shows a related but distinct trajectory: declining from **8.6%** (Colonial) to **3.0%** (Oligarchic), then partially recovering to **4.7%** (Modern), consistent with twentieth-century transnational re-integration of elites through diplomacy, business, and migration.

4.5.5 Fragility

Figure 4 contrasts targeted removal (ordered by betweenness-based criticality) with random removal: the targeted trajectory collapses the largest connected component after very few deletions, whereas random removal produces a nearly linear decline.

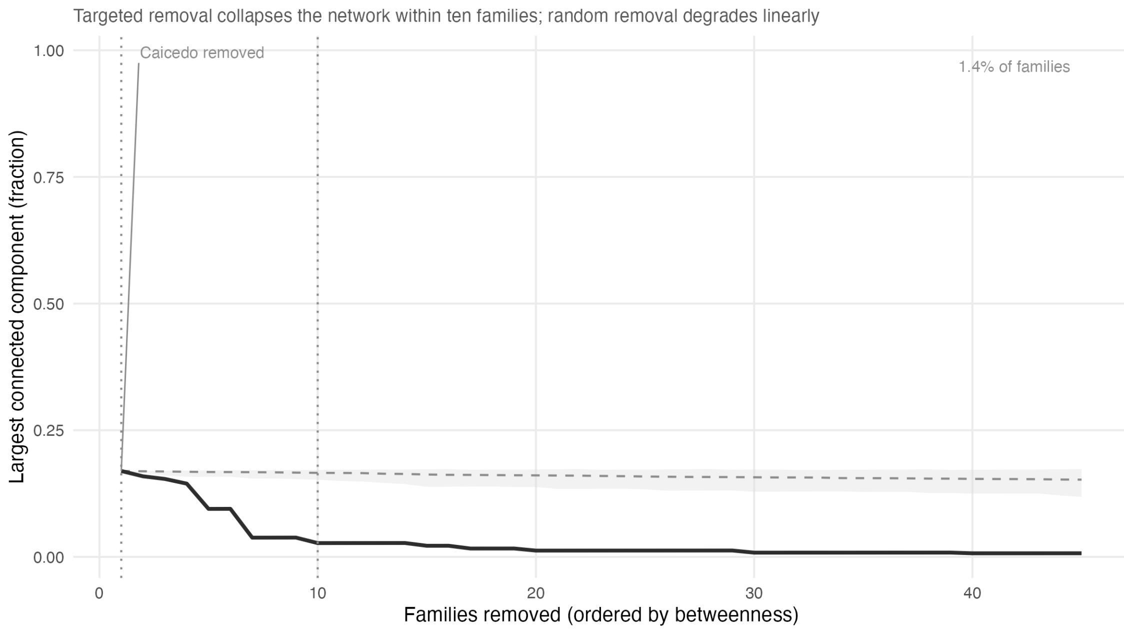

Figure 4. Largest connected component under targeted versus random family removal.

Targeted removal of highly central families produces rapid network fragmentation, in contrast to the gradual degradation observed under random removal. A very small number of removals is sufficient to collapse the largest connected component.

Families identified as structurally critical in this exercise overlap substantially with those in the dense core, indicating convergence between latent structure and network robustness. This convergence is partly mechanical — both the SBM and the betweenness-ordered attack reward families with high connectivity — but the specific overlap (six of the top ten critical families are SBM-core members, out of 21 core families in a corpus of 743) exceeds what random block assignment would produce. The core is therefore not only dense but also functionally central to connectivity.

4.5.6 A robust core: four Medici-like brokers

The structural tension identified in the brokerage subsection above has an empirical resolution in a small set of families that occupy the core, carry substantial brokerage load, and appear among the critical removal targets. Operationalizing “robust brokers” by the intersection of three criteria — SBM core membership, top-decile aggregate brokerage, and inclusion in the top thirty critical removal targets — returns four families: **Caicedo** (Colombia), and **Cruz**, **Pinto**, and **Vicuña** (Chile). These four are the only families in the corpus that combine dense embedding in the elite core with extensive bridging to the periphery (Figure 5).

The typological reading follows Padgett and Ansell's (1993) analysis of Medici robustness. In their account, Medici dominance in Renaissance Florence was not a matter of being *the most connected* family (they were not) but of being densely embedded in a cohesive core while simultaneously brokering across factions. That dual position generated *robust action*: the capacity to act decisively in one domain without compromising flexibility in another. Caicedo, Cruz, Pinto, and Vicuña occupy the same structural position in the Latin American kinship graph. Their brokerage role profile is consistent with the typological match: across these four families, gatekeeper and representative roles (brokerage *across* block boundaries) account for roughly **23%** of aggregate brokerage, against roughly **11%** for a contrast class of high-brokerage peripheral families (García-Mansilla, Echaurren, Ospina, Arboleda, Chávez, Balcarce, Podestá, Onassis). Because the contrast class is small and purposively selected — the highest-brokerage peripheral families by aggregate score — the comparison should be read as a structured illustration rather than as a hypothesis test; a configuration-model baseline is priority work for a follow-up paper. Within that caveat, Caicedo-type families broker between blocks at approximately twice the rate of peripheral brokers, who predominantly coordinate *within* the periphery.

The contrast class matters substantively. García-Mansilla (Argentina), Echaurren (Chile), and Ospina (Colombia) have *higher* absolute brokerage totals than Caicedo or Pinto, but they sit outside the structural core. They are dense local coordinators — presidents' brothers and cousins, ministerial networks, provincial elites — whose removal does not fragment the graph because their ties are redundant within their own block. The Medici-like families combine *scarcer* structural position with *rarer* role diversity; the counter-cases combine *denser* local embedding with *narrower* role specialization.

The finding that three of the four robust brokers are Chilean is consistent with the national composition of the SBM core (16 of 21 core families are Chilean) and with the negative degree–closure correlation documented under H1 (**Corr = −0.536** for Chile, the least negative among large republics — reflecting that Chilean core families are both high-degree *and* moderately endogamous, unlike Mexican or Argentine cases where very high degree accompanies very low endogamy). Caicedo's Colombian exceptionalism in this set is worth flagging as a direction for archival follow-up: if the Caicedos of the Cauca Valley occupy a structurally Medici-like position in a polity that otherwise lacks a Chilean-style core, that is either a finding about Caicedo or a finding about Colombia, and only archival work can distinguish the two.

4.5.7 Summary of H5

Table 10. Summary of H5 evidence across analytical approaches.

Analysis	Diagnostic	Finding	Corroborated by
ERGM	nodematch.pais = 1.78 (OR 5.94)	Sparse but sorted tie formation	Homophily H(Country) = 0.48
Homophily	H(Tier) = 1.00; H(Office) = -0.86	Tier closure + functional complementarity	ERGM absdiff.mean_degree < 0
SBM	K = 2; pi_22 / pi_11 = 208	Core-periphery: 21 vs 722 families	Attack: 6 of top 10 are core
Temporal	Cross-family share: -70% across 5 cohorts	Monotonic oligarchic consolidation	Cross-country share also falls
Attack	LCC to 17% after 1 removal	Fragility to targeted removal	SBM core membership
Brokerage	Cross-block share: core 23%, periphery 11%	Four Medici-like robust brokers	A3 core + A6 top decile + A10

No single analysis is decisive on its own. The contribution of H5 lies in the convergence across **complementary** approaches. The six analyses draw on different graph representations (family-level kinship graph, multiplex graph with institutional layers, cohort-sliced subgraphs), different inference procedures (maximum-likelihood ERGM, variational SBM, permutation-based homophily tests, betweenness-ordered attack simulation, Gould–Fernandez role decomposition), and different null models (edges-only baseline, random block assignment, random removal order). These are complementary perspectives on a common underlying graph, not statistically independent tests: they share the same raw kinship edges and can therefore respond in partially correlated ways to the same data features. What distinguishes them is the diversity of inferential assumptions each brings to bear — a latent-block view, a tie-formation view, a homophily view, a temporal view, a robustness view, a brokerage view. The substantive contribution of H5 is that these heterogeneous assumptions converge on the same architectural description: sparse but sorted, historically consolidated, Chilean-dominated at the core, and dependent on a small set of robust brokers for connectivity beyond the core. No single analysis, taken alone, would license this claim with comparable confidence.

H5 receives strong descriptive support based on the agreement across these complementary indicators.

5 Discussion and conclusion

5.1 Heterogeneity versus a single Latin American model

Our results are consistent with a **heterogeneous Latin America**: elite kinship structures are not isomorphic across countries. Contrasting chain-like configurations in Paraguay with dense or multi-hub structures in Mexico, Colombia, or Argentina aligns with institutionalist accounts of divergent state formation. These differences emerge even when all cases are measured through a single digital corpus, suggesting that the variation reflects underlying historical processes rather than purely methodological artifacts.

5.2 Chilean exceptionalism as structural pattern

H5 raises a finding that the heterogeneity frame does not fully accommodate: **sixteen of the twenty-one families in the SBM core, and three of the four robust brokers,**

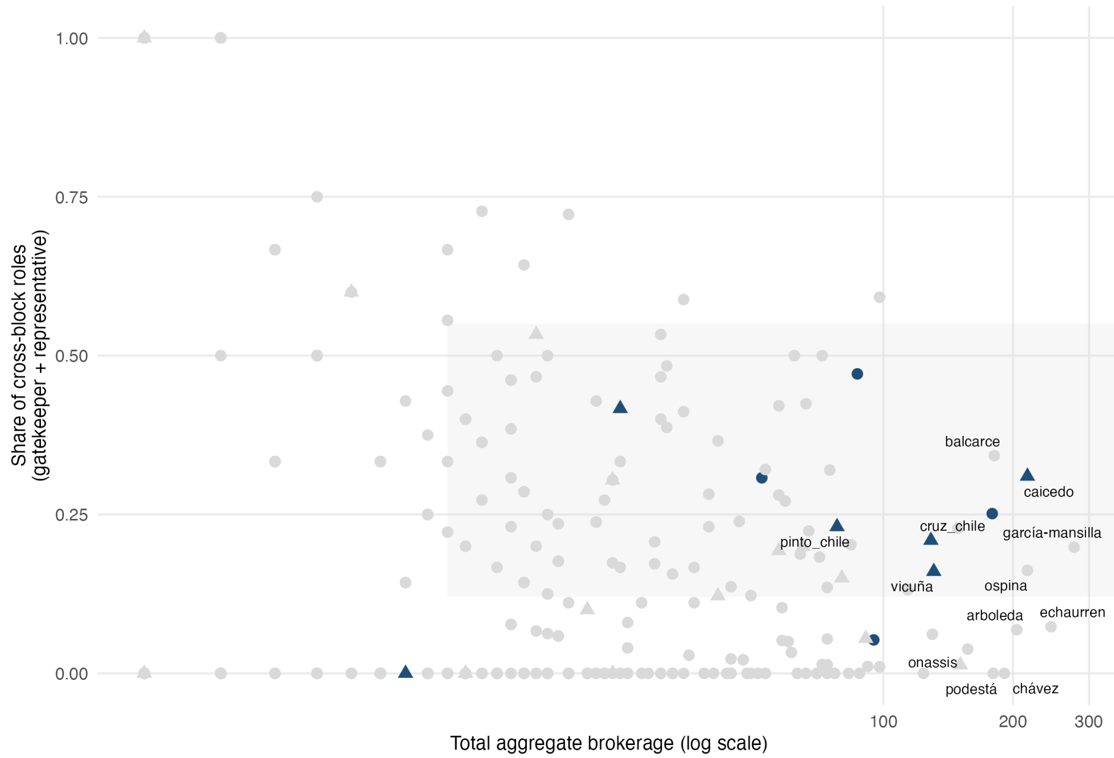

Figure 5. Robust-core families (intersection of SBM core, top-decile aggregate brokerage, and top-30 critical removal targets) versus counter-case peripheral brokers. Triangles: SBM core families. Circles: periphery. Dark fill: families in the top 30 of the attack simulation. Labeled points: the four robust brokers (Caicedo, Cruz, Pinto, Vicuña) and selected counter-cases.

are Chilean. This is not a minor imbalance. Chile contributes **99** families to the corpus — fewer than Argentina (**161**), Peru (**185**), or Colombia (**146**) — yet dominates the structurally central positions. The pattern is not reducible to sub-corpus size; if anything, the opposite holds.

Two readings are available, and we do not adjudicate between them here.

As a structural fact. Chilean elite continuity across the Portalian, Parliamentary, Popular Front, and early neoliberal regimes is a standing puzzle in Latin American political sociology. Zeitlin (1984), Stabili (1996), and Valenzuela and Valenzuela (1976) variously attribute this continuity to (i) a restricted electoral regime from 1833 to the 1920s that insulated elite marriage markets from mass politics, (ii) the absence of a successful land reform until 1967 that preserved hacendado lineages, and (iii) Santiago's unusual demographic dominance within Chile, concentrating the marriage market geographically in ways that Buenos Aires, Lima, or Mexico City — each rivaled by strong regional capitals — did not. The SBM core is consistent with these accounts: it identifies precisely the structural signature (densely interconnected small group, peripheral brokerage layer) that a closed Santiago elite would produce.

As a sampling artifact. Spanish-language Wikipedia editorial practices vary across countries. If Chilean editors are disproportionately attentive to genealogical detail — recording spouses, parents, and children more consistently than Peruvian or Argentine editors — then Chilean families would appear more densely linked not because they *are* more densely linked, but because their links are more completely documented. The ratio of matched to total kinship rows (**57.8%**, Section 3) is a global figure; country-level variation in this ratio would partially test the artifact hypothesis. A first-order diagnostic — comparing per-country edge-matching rates — is feasible within the existing pipeline and is a priority for the next iteration of the project; until that test is performed, the artifact reading cannot be excluded.

The two readings are not mutually exclusive, and both generate testable predictions that future work should pursue: country-specific edge-matching rates, archival cross-validation of core-family genealogies, and comparison against non-Wikipedia elite registers (where available) for Chile, Colombia, and one contrast country. The characterization of the Chilean-dominated core as a **structural pattern in the documentary elite field** should, for now, be read with this caveat attached: it is a robust feature of the corpus as constructed, and a candidate fact about Latin American elite reproduction whose corpus-external validation remains open.

5.3 The Pacific corridor

The Pacific–Andean emphasis in cross-border weights (Table 3) is not reducible to geographic proximity alone. Bolivia and Ecuador are proximate to several large republics yet do not dominate the top dyads in this corpus; the pattern therefore aligns more closely with **historical administrative and biographical pathways** than with a simple distance-decay model. Viceregal structures once centered on Lima linked Andean and coastal spaces into a common political field; the War of the Pacific (1879–84) further entangled Chilean and Peruvian elites through conflict, exile, and subsequent documentation in biographical fields.

The kinship/biographical decomposition matters here. For Chile–Peru, **69%** of counted events are biographical (**81** of **117**) and **31%** kinship (**36** of **117**), which may partly reflect Chilean encyclopedic attention to Peruvian figures and exiles after 1879, not only cross-border genealogy. By contrast, Argentina–Chile is predominantly kinship-weighted (**60%** kinship, **40%** biographical on **94** events), consistent with denser genealogical entanglement including documented migration and political exile corridors. Argentina–Uruguay is mixed (**44%** kinship, **56%** biographical on **105** events), consistent with a shared Rioplatense cultural field in which family and professional circulation are both visible. The three pairs thus carry three different relational signatures, even though they rank close together on total weight.

The absence of Argentina–Brazil among the strongest pairs is the strongest implicit test available for the administrative-corridor hypothesis: if geographic proximity alone drove the pattern, the two largest adjacent economies should dominate the matrix. They do not. This pattern is consistent with colonial administrative legacies and Rioplatense cultural circuits structuring elite ties more than uniform neighborhood effects across Latin America — an empirical regularity that future comparative work on Brazilian elite networks (using Portuguese-language sources) could extend or challenge.

5.4 Oligarchic consolidation

H5's temporal analysis (Table 9) documents a monotonic **70%** decline in the cross-family tie share from the Colonial cohort to the Modern cohort, coinciding with rising total edge volume through the Republican period. This conjunction — more ties, fewer lineages — is the empirical signature of oligarchic consolidation, and it occurs during the period Centeno (2002) identifies as the formative century of the Latin American state (roughly 1820–1920). Centeno's argument is that Latin American republics did *not* undergo the fiscal-military transformation that Tilly (1990) and others associate with European state formation, because they did not fight the mass wars that forced such transformations in Europe. Instead, Centeno argues, they produced **"states without wars"**: administrative structures sustained not by centralized fiscal capacity but by elite network coordination.

The present data do not test Centeno's mechanism directly. What they show is consistent with its structural prediction: if Latin American states consolidated through kinship intensification rather than bureaucratic institution-building, the network signature should be precisely the pattern we observe — rising edge volume in the state-building cohorts, concentrated within a shrinking set of structurally dominant lineages, producing a core–periphery architecture by the end of the consolidation period. Whether this structural signature *caused* the institutional weakness Centeno describes, or merely accompanied it, is a question for designs that exploit variation in the timing of elite consolidation across countries — another priority for cohort-level follow-up work.

5.5 Alternative explanations

Four alternative explanations warrant treatment before the observed patterns can be read as informative.

Wikipedia visibility bias is the most serious. If some families are documented in Wikipedia simply because they are more famous — rather than because they are more networked — then high degree could reflect celebrity rather than structural position. Two observations partially mitigate this concern. First, the negative correlation between degree and endogamy (H1) would not follow from a pure visibility story: celebrity families are not systematically more exogamous than obscure ones. Second, the topological variation across countries — chain in Paraguay, star in Chile, web in Argentina — is not reducible to differential Wikipedia coverage, since all countries are drawn from the same Spanish-language corpus under comparable extraction rules. Visibility bias would inflate all graphs equally; it cannot explain *differential* topology. It can, however, still explain *which families* dominate within each graph — a point that recurs in the Chilean-exceptionalism discussion above.

Demographic structure offers a second alternative: countries with larger populations should produce denser networks simply by having more nodes. Mexico and Argentina are the most populous countries in the corpus and do show dense webs — but Chile, with fewer than half Mexico's population, also shows a dense star structure, while Bolivia and Ecuador, with smaller corpora, show fragmentation rather than simple sparsity. The topology classification is not monotonically predicted by corpus size. Paraguay's linear chain emerges from the smallest national corpus ($n = 21$), which is consistent with patrimonial consolidation rather than with any density artifact.

Random network formation is the null hypothesis that H1–H4 treat interpretively but H5 tests formally. The ERGM in Table 6 rejects an edges-only baseline with high confidence: `nodematch.pais` $p < 10^{-40}$ and `nodecov.n_members` $p < 10^{-115}$ under MCMC maximum likelihood estimation. The observed sorting by country and lineage size is not produced by sparse random graphs with the same edge count. An Erdős–Rényi random graph with the same degree sequence as the Chilean subgraph would not produce a star topology; star formation requires a generative process (preferential attachment, or a founding lineage with historically higher documentation probability) that is itself non-random. Configuration-model baselines for the SBM partition — not estimated here — would further strengthen the core–periphery claim and remain priority work for a follow-up paper.

Core–periphery as random clustering. Any sparse graph will exhibit some unevenness in density. The question is whether the SBM's detected core is statistically distinguishable from what a null model would produce. Two facts argue that it is. First, the within-core tie probability (0.138) is three orders of magnitude above the periphery's within-block probability (0.00066) and two orders of magnitude above the cross-block probability (0.0020). Structural equivalence at this scale is not produced by sparse random graphs with the observed edge count. Second, the convergence between SBM core membership and attack-simulation criticality — six of ten critical families are core members — is not predicted by any null model that treats the blocks as interchangeable. Together these facts make the core–periphery structure a fact about the graph, not an artifact of the detection method.

5.6 Wikipedia as archive

Reframing Wikipedia as an *archive of collective documentation* clarifies the epistemic status of this study: we analyze who is repeatedly named and linked in a public archive — not the full elite stratum of each society. This reframing is more than a caveat. It converts the sample-selection problem into a substantive object: the *documentary elite field* is what we study, and its boundaries are themselves sociologically informative. Who gets inscribed in Wikipedia — and who does not — reflects decisions made by dispersed editors over two decades, shaped by language, political orientation, media coverage, academic attention, and family memory. The documentary elite field is therefore not a proxy for the social elite; it is a distinct object whose structure (as documented here) bears systematic relation to the social elite but is not identical with it.

This move aligns with computational social science cautions about digital trace data (Lazer et al., 2014) and with calls for transparency in algorithm-assisted history.

5.7 Concluding remarks

Drawing on Wikipedia-derived kinship networks for **6,711** Latin American notables and **8,201** matched kinship edges (embedded in a larger multiplex graph for centrality), this study documents substantial cross-national heterogeneity in elite family structure. Descriptive patterns align directionally with five working hypotheses — closure and concentration, topology and stylized regime narratives, Pacific-leaning transnational weights, identifiable betweenness brokers, and the integrative core–periphery architecture under H5 — while remaining strictly observational. The most robust takeaway at the level of H1–H4 is **diversity on surface features**: elite kinship in Latin America does not reduce to a single network form at the level of country topology.

H5 reframes the contribution. The surface findings (H1–H4) describe regularities in an elite corpus; the deeper finding is that these regularities share a generative substrate. Latin American elite networks, as visible through Spanish-language Wikipedia, are organized by a sparse-but-sorted tie-formation process (ERGM), sort by structural tier and country while specializing by political office (homophily), partition into a Chilean-dominated dense core and a large sparse periphery (SBM), consolidate this structure through nineteenth- and early-twentieth-century cohorts (temporal analysis), and depend on a small number of Medici-like families — **Caicedo**, **Cruz**, **Pinto**, **Vicuña** — to mediate between the core and the rest of the graph.

The convergence of six **complementary** analyses on the same structural object is the paper's strongest epistemic claim; see Table 10 for a consolidated view. These analyses are not statistically independent — they draw on a common underlying graph — but they deploy different inferential assumptions: an ERGM fit could be unstable, an SBM could over-partition, an attack simulation could be noisy, a temporal split could be cohort-sensitive, a brokerage decomposition could be sensitive to role-coding assumptions, a homophily index could be inflated by attribute granularity. Jointly, they tell a single story. The Latin American documentary elite, in this corpus, is not a scatter of national networks loosely connected by corridors. It is a single architecture with a Chilean-dominated core, a documented history of consolidation, and an identifiable set of robust brokers — Caicedo, Cruz, Pinto, and Vicuña — holding it together. Whether that architecture reflects the social elite itself, or the specific way this elite has been inscribed in Spanish-language Wikipedia, is a question our data position us to pose sharply but not yet to answer definitively.

We close with two implications that extend beyond Latin American historiography. First, *digital genealogies increasingly shape public imaginaries of elite connection.* Scholarly use of such sources demands methodological transparency and humility about whose ties get documented and whose remain invisible — a point that applies to any Wikipedia-based prosopography, not only this one. Second, *the documentary elite field is increasingly legible to non-scholars.* Journalists, activists, and political opponents can now cross-reference kinship with corporate boards, public contracts, and party donations at a scale that was unthinkable a generation ago. The families that inscribed themselves most densely in Wikipedia are, by construction, the families most exposed to network-based accountability journalism. Where network closure and institutional concentration coincide — as they do in several of the polities analyzed here — the transparency afforded by open genealogical data may paradoxically reinforce elite salience by making visible precisely those connections that elite families have long relied upon but rarely had to defend publicly. Understanding the structure of the documentary elite field is therefore not merely an academic exercise: it is a prerequisite for assessing the political consequences of open genealogical data.

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